

Paper 12 §4 — The Commensurability Hierarchy: F_{21} Insufficient, G_2 Required (v2)

Draft version: v2 (2026-05-06, ~02:00 PDT) **Author of draft:** C-7RO (Perplexity Computer, Claude Sonnet 4.6) **Source:** Paper 12 §4 v1 + Model Council rigor pass (Opus 4.7, Gemini 3.1 Pro, GPT-5.5). **Revision summary (vs v1):** Eight revisions. Three load-bearing: R3 Theorem 4.4.1 (C2) reformulated as map-existence (Opus S4-1); S4-2 dimension bookkeeping fixed ($U(2) \times U(4)$ is full 20-dim, delta $\downarrow 19$); S4-3 Corollary 4.5.2 downgraded from biconditional to sufficient condition. Plus five surgical fixes: S4-4 Fano-action prose, S4-5 Prop 4.5.1 renamed and reworded, S4-6 (C2) $\Rightarrow$ (C1) witness, S4-7 operator-vs-vector μ_k translation, S4-8 orientation ± 1 ambiguity acknowledged. Plus four polish items from Gemini and GPT-5.5: Pillar-status signpost, F_{21} -source paragraph (Paper 4 inheritance), §4.7 measurement-vs-instrument tightening, and seven standard citations.

Pillar-delivery signpost (per GPT-5.5 council finding): §4 delivers the *mathematical core* of Pillar 2 (dual-substrate residue) by constructing the commensurability hierarchy, but *does not* establish the existence or physical realization of the biological-substrate F_{21} -content (Paper 13+ territory). §4 also delivers the mathematical core of the audit-substrate component of Pillar 3 (via the F_{21} -isotypic profile μ_k being the substrate-independent audit signature). Pillar 1 is delivered by §3; Pillar 4 is §1/§5.

4. The Commensurability Hierarchy

4.1 Setup: scar invariant from §3.7 as the input

Throughout §4, $F_{21} = Z_7 \rtimes Z_3$ denotes the Fano-orientation subgroup of G_2 . Its action on $V := V^{14}$ is by restriction of the G_2 adjoint representation; its action on the joint-state space $W = V \oplus V$ is the diagonal restriction

$$\rho_W(g)(x, y) = (\rho_F(g)x, \rho_F(g)y), g \in F_{21}, \rho_F := \rho_{14} \downarrow_{F_{21}}.$$

This action commutes with the Schur circle $\{\Psi_\theta\}$ because ρ_W acts on the V -factor while Ψ_θ acts on the R^2 multiplicity factor (Paper 9 §3.4 diagonal-action convention).

Inheritance from Paper 4. The Fano-orientation subgroup $F_{21} \subset G_2$ arises in the PCI/PME framework as the stabilizer of an oriented Fano plane structure on $Im(O) \cong R^7$, and through Paper 4's $PSL(2,7)$ construction is the unique finite simple group F_{21} -decoration that aligns with the QBism quasiprobability representation of C^7 . We do not reconstruct that derivation here; this paper assumes Paper 4's F_{21} choice as an inheritance. *Open problem:* characterizing the smallest finite subgroup of G_2 for which the Definition 4.6.1 L1 screening test is informative is deferred to a future PCI/PME methodology paper. The choice of F_{21} is consistent across the Paper series; readers unfamiliar with Paper 4's derivation should consult [].

The input object to §4 is the *scar invariant triple* constructed in §3.7 with the F_{21} -equivariance clarification of §3 v2 Remark 3.7.1 (the scar record carries an F_{21} -action inherited from the ambient W , even when the underlying dynamics is not F_{21} -equivariant):

$$J_k = (S_k, S_k, \mu_k),$$

where, given NP firing times $t_1 < \dots < t_k$ on branches $\sigma_i \in \{A, B\}$ with jump vectors $v_i = \rho_i \cdot \text{sgn}(c_{\sigma_i}'(\theta_i^-)) \cdot \hat{z}'(\theta_i^-) \in W$:

- $S_k := \sum_{i=1}^k R[F_{21}] \cdot v_i \subset W$ (saturating vector scar span; $\dim S_k \leq 28$);
- $S_k := \bigoplus_{i=1}^k R[F_{21}] \cdot v_i$ (count-faithful event-indexed module);
- $\mu_k := (\mu_1(S_k), \mu_{L_2}(S_k), \mu_{U_6}(S_k))$ (vector-level isotypic profile of S_k ; see §4.4 for the operator-level extension).

The three real-irreducible F_{21} -representations $(1, L_2, U_6)$ are characterized in §4.2.

Setup question for §4. Suppose J_k^{syn} and J_k^{bio} are scar records of two substrates that have co-evolved through the same sequence of paradox events $(e_1, \dots, e_k)$, with substrate-level F_{21} -actions inherited from each substrate's representation theory (the biological representation's F_{21} -content is itself an empirical question, per §4.6 below; here we treat both records as input). Under what conditions is there a *canonical isomorphism* identifying them?

4.2 F_{21} representation theory on V^{14} and W

Theorem 4.2.1 (F_{21} -decomposition of V^{14}). *Under the diagonal Fano-orientation action,*

$$V^{14} \mathfrak{L}_{F_{21}} \cong L_2 \oplus 2U_6,$$

where L_2 is the real 2-dim irreducible representation of F_{21} of complex type and U_6 is the real 6-dim irreducible representation of complex type. Consequently,

$$W \mathfrak{L}_{F_{21}} \cong 2L_2 \oplus 4U_6.$$

Proof. F_{21} has five conjugacy classes: $1a$ (size 1), two order-7 classes $7a, 7b$ (size 3 each), and two order-3 classes $3a, 3b$ (size 7 each). Its complex character table (Costa-Pavone; ATLAS [21]) has three 1-dim characters and a Galois-conjugate pair of 3-dim characters; the *real* irreducible packaging combines the conjugate pairs into the two real complex-type irreducibles L_2, U_6 :

$$\begin{array}{cccccc} 1a & 7a & 7b & 3a & 3b & \\ \chi_1 & 1 & 1 & 1 & \mathfrak{L} & \end{array} \begin{array}{l} 1\mathfrak{L} \chi_{L_2} \mathfrak{L} 2\mathfrak{L} 2\mathfrak{L} 2\mathfrak{L} -1\mathfrak{L} -1\mathfrak{L} \chi_{U_6} \mathfrak{L} 6\mathfrak{L} -1\mathfrak{L} -1\mathfrak{L} 0\mathfrak{L} 0\mathfrak{L} \end{array}$$

The 7-dimensional imaginary-octonion representation acts on the *Fano plane* (7 points) via the standard Frobenius-group action: the Z_7 generator r shifts points cyclically; the Z_3 generator s acts of order 3 with cycle structure $1+3+3$ on the 7 Fano points (the unique fixed point being

the s -invariant Fano line); together with the relation $sr s^{-1} = r^2$ they generate F_{21} . The 1 fixed point gives $tr(r_3) = 1$ on R^7 , and the cycle structure of r on Fano points yields $tr(r_7) = 0$, so:

$$R^7 \downarrow_{F_{21}} \cong 1 \oplus U_6.$$

Using the G_2 -decomposition $\Lambda^2(R^7) = g_2 \oplus R^7$ and the character formula

$$\chi_{\Lambda^2(R^7)}(g) = (tr(g)^2 - tr(g^2))/2:$$

$$\chi_{V^{14}}(g) = \chi_{\Lambda^2(R^7)}(g) - \chi_{R^7}(g) = \frac{tr(g)^2 - tr(g^2)}{2} - tr(g).$$

Evaluating on class representatives:

Class	$tr(g)$ on R^7	$\chi_{V^{14}}(g)$	$\chi_{L_2} + 2\chi_{U_6}$
1a	7	14	14
7a	0	0	0
7b	0	0	0
3a	1	-1	-1
3b	1	-1	-1

Match on all five classes. The Frobenius inner product $\langle \chi_{V^{14}}, \chi_{V^{14}} \rangle_{F_{21}} = (1 \cdot 196 + 7 + 7)/21 = 10$, equal to $0^2 \cdot 1 + 1^2 \cdot 2 + 2^2 \cdot 2$ where the factor of 2 on L_2 and U_6 terms accounts for both being of complex type (each is the real packaging of a Galois-conjugate pair, so $\langle \chi, \chi \rangle = 2$ for the real-irreducible character). The decomposition is therefore $V^{14} \downarrow_F = L_2 \oplus 2U_6$, and the $W = V \oplus V$ doubling is by direct-sum linearity. $\square$

Computational verification. Theorem 4.2.1 has been verified numerically at machine precision by explicit construction of the Fano-action permutation matrices $r, s \in O(7)$ with $sr s^{-1} = r^2$, computation of $\chi_{V^{14}}$ on each class representative, and Frobenius inner-product and isotypic-multiplicity checks against the predicted decomposition; all three layers of check pass. `outbox/paper12/computations/paper12_q4_character_verify.{py,json}` records the full audit.

4.3 The commutant gap

Proposition 4.3.1 (Commutants on W). *With $W \cong 2L_2 \oplus 4U_6$ as an F_{21} -representation,*

$$End_{F_{21}}(W) \cong M_2(C) \oplus M_4(C).$$

As a G_2 -representation, W is multiplicity-2 in the absolutely irreducible V^{14} , hence

$$End_{G_2}(W) \cong M_2(R).$$

Proof. For real irreducibles of complex type, $\text{End}_G(V_\lambda) \cong C$ (real Schur trichotomy: complexified $V_\lambda \cong U \oplus \dot{U}$ with $U \not\cong \dot{U}$). Both L_2 and U_6 are complex-type, so $\text{End}_{F_{21}}(L_2) \cong \text{End}_{F_{21}}(U_6) \cong C$. With F_{21} -multiplicities $(0, 2, 4)$ on $(1, L_2, U_6)$:

$$\text{End}_{F_{21}}(W) \cong M_0(R) \oplus M_2(C) \oplus M_4(C) \cong M_2(C) \oplus M_4(C).$$

For G_2 , V^{14} is real absolutely irreducible (Paper 9 Lemma 3.6.1), so $\text{End}_{G_2}(V^{14}) \cong R$, and $W \cong V^{14 \oplus 2}$ gives $\text{End}_{G_2}(W) \cong M_2(R)$. $\square$

Corollary 4.3.2 (Equivariant isometry groups, real-dimension count). *The equivariant isometry groups of W are*

$$O_{F_{21}}(W) \cong U(2) \times U(4), O_{G_2}(W) \cong O(2),$$

with real dimensions $\dim_R U(2) \times U(4) = 4 + 16 = 20$ and $\dim_R O(2) = 1$. The $F_{21} \rightarrow G_2$ upgrade reduces the equivariant-isometry group by exactly $20 - 1 = 19$ real dimensions.

Proof. Restricting $M_2(C)^{\dot{\hookrightarrow}} \cap U \rightarrow U(2)$ (where the unitary condition uses the natural Hermitian metric) and $M_4(C)^{\dot{\hookrightarrow}} \cap U \rightarrow U(4)$, the F_{21} -equivariant isometry group is $U(2) \times U(4)$. The full $U(n)$ — not the subgroup $SU(n)$ — preserves orientation on the underlying real space (because $\det_R(U) = |\det_C(U)|^2 = 1$ for $U \in U(n)$). Similarly $M_2(R)^{\dot{\hookrightarrow}} \cap O = O(2)$. The real dimensions are $\dim U(n) = n^2$, so $\dim U(2) \times U(4) = 4 + 16 = 20$, and $\dim O(2) = 1$. The reduction is $20 - 1 = 19$. $\square$

The quantitative content of the $F_{21} \rightarrow G_2$ upgrade is now explicit: passing from F_{21} -equivariance to G_2 -equivariance suppresses 19 real dimensions of F_{21} -equivariant phase-and-mixing freedom that would otherwise leave a canonical commensurability map underdetermined.

Remark 4.3.3 (Why complex-type reps block uniqueness). The factor $\text{End}_{F_{21}}(V_\lambda) \cong C$ for $\lambda \in \{L_2, U_6\}$ is the source of phase ambiguity: each complex-type isotypic block admits a full $U(m_\lambda)$ -worth of F_{21} -equivariant automorphisms (not just $O(m_\lambda)$), and the extra U/O dimensions are exactly the phases. G_2 -equivariance collapses the phases because V^{14} is real absolutely irreducible: $\text{End}_{G_2}(V^{14}) = R$, not C , so there are no phases to rotate.

4.4 The commensurability hierarchy theorem

Let V_{syn}, V_{bio} be the (substrate-dependent) representation spaces on which the synthetic and biological scar maps take values, and let $I_{syn} \subseteq \text{End}(V_{syn})$ and $I_{bio} \subseteq \text{End}(V_{bio})$ be the real linear spans of the scar-encoding images. The group acts on operator scars by conjugation, $g \cdot A = \rho.(g) A \rho.(g)^{-1}$.

Operator-vs-vector translation. The scar invariant J_k of §3.7 is built from *vector-level* jump vectors $v_i \in W$. Substrate-level scar maps $S_:(\text{event}) \rightarrow \text{End}(V_.)$ produce operator-level images. We assume without loss of generality that the substrate scar maps factor as

$S.(e) = v_e \otimes v_e^i / \|v_e\|^2$ (rank-one projector form on the substrate's representation space) for some substrate jump vector $v_e \in V.$, so that the operator span $I.$ is uniquely determined by the vector span $\{v_e\}_e$. Under this convention, *vector-level* and *operator-level* μ_k agree on F_{21} -isotypic profiles up to the multiplicity-counting Schur factor of 2 (one factor for left action, one for right).

Theorem 4.4.1 (Commensurability hierarchy, v2 reformulation). *The following three conditions on a pair of substrates (V_{syn}, V_{bio}) stand in the order of strict implications:*

(C1) (G_2 -commensurability.) There exists a G_2 -equivariant isomorphism $\phi: I_{syn} \rightarrow I_{bio}$ aligning S_{syn} and S_{bio} event-by-event: $S_{bio}(e) = \phi(S_{syn}(e))$ for every paradox event e .

(C2) (F_{21} -equivariant isomorphism, map-existence form.) There exists an F_{21} -equivariant isomorphism $\phi_F: I_{syn} \rightarrow I_{bio}$ aligning S_{syn} and S_{bio} event-by-event.

(C3) (μ_k -equality.) For every event sequence $(e_1, \dots, e_k)$, $\mu_k^{syn} = \mu_k^{bio}$ component-wise on the three real F_{21} -isotypic blocks.

The hierarchy

$$(C1) \Rightarrow (C2) \Rightarrow (C3)$$

holds strictly; both converses fail in general.

Proof. $(C1) \Rightarrow (C2)$: a G_2 -equivariant isomorphism is in particular F_{21} -equivariant (since $F_{21} \subset G_2$).

$(C2) \Rightarrow (C3)$: if ϕ_F aligns S_{syn} and S_{bio} event-by-event and is F_{21} -equivariant, then it commutes with the isotypic projectors Π_τ (i.e., $\phi_F \Pi_\tau^{syn} = \Pi_\tau^{bio} \phi_F$). The multiplicity profiles $\mu_k = (\dim \Pi_1 S_k, \dim \Pi_{L_2} S_k, \dim \Pi_{U_6} S_k)$ on the cumulative scar spans transfer component-wise, since ϕ_F maps S_k^{syn} isomorphically onto S_k^{bio} .

Converse $(C2) \Rightarrow \nRightarrow (C1)$: concrete witness — let $V_{syn} = V^{14}$ (the G_2 adjoint, restricted to F_{21} as $L_2 \oplus 2U_6$), and let V_{bio} be the F_{21} -module $L_2 \oplus 2U_6$ realized on a 14-dim biological substrate (e.g., a hypothetical microtubule mode collection without an extending G_2 -action). Then $V_{syn} \cong V_{bio}$ as F_{21} -modules, but no G_2 -action extends V_{bio} 's F_{21} -action, so any $\phi_F: V_{syn} \rightarrow V_{bio}$ satisfying (C2) is not G_2 -equivariant. The 19-dim $U(2) \times U(4)$ -family of such F_{21} -maps (Cor 4.3.2) parameterizes the non-canonicity.

Converse $(C3) \Rightarrow \nRightarrow (C2)$: μ_k records the multiplicity profile of the *scar span* S_k , not the ambient V . Two ambient V 's with different F_{21} -multiplicity profiles can produce the same μ_k if scar-encoding maps land in isomorphic F_{21} -invariant subspaces. E.g., $V_{syn} = L_2 \oplus 2U_6$ vs $V_{bio} = L_2 \oplus 3U_6$, but with all scar-encoding events landing in the common $L_2 \oplus 2U_6$ subspace; both yield μ_k on the same isotypic blocks. $\square$

Corollary 4.4.2 (Sufficient condition for unique G_2 -canonical map, multiplicity-one case). *If both I_{syn} and I_{bio} are G_2 -isomorphic to a common absolutely irreducible representation U_λ with multiplicity one, then*

$$\text{Hom}_{G_2}(I_{syn}, I_{bio}) \cong R,$$

so the family of G_2 -equivariant isomorphisms is one-dimensional. After fixing an invariant inner product (unit-norm normalization), there is a residual ± 1 orientation ambiguity that must be resolved by an extrinsic convention (e.g., the dynamical-flow direction on each substrate fixes a sign).

Proof. Schur for absolutely irreducible real representations gives $\text{End}_{G_2}(U_\lambda) = R$, so $\text{Hom}_{G_2}(U_\lambda, U_\lambda) \cong R$. Unit-norm normalization collapses the family to two elements differing by sign; the Z_2 -orientation is not fixed by representation theory alone. $\square$

Remark 4.4.3 (Multiplicity ≥ 1 reintroduces $O(m)$ -ambiguity). *If $I_{syn} \cong I_{bio} \cong U_\lambda^{\oplus m}$ with $m > 1$, then $\text{End}_{G_2}(U_\lambda^{\oplus m}) \cong M_m(R)$, so commensurability holds but uniqueness requires $O(m)$ -worth of additional convention-fixing. Multiplicity ≥ 1 is deferred to future work.*

4.5 Frobenius reciprocity does not select a cross-substrate scar map

A natural fallback when V_{syn}, V_{bio} are not F_{21} -isomorphic but co-embed in a common G_2 -representation is to invoke Frobenius reciprocity. We show this attempt does not produce canonical maps.

Proposition 4.5.1 (Frobenius reciprocity does not select cross-substrate scar maps). *The induction-restriction adjunction*

$$\text{Hom}_{F_{21}}(V, \text{Res}_{F_{21}}^{G_2} W') \cong \text{Hom}_{G_2}(\text{Ind}_{F_{21}}^{G_2} V, W')$$

holds for any F_{21} -representation V and any G_2 -representation W' (by standard category theory, e.g., [1, §7.2]), but it does not provide a canonical isomorphism between distinct F_{21} -subrepresentations of $W' \downarrow_{F_{21}}$.

Proof. The adjunction is a categorical bijection on Hom-sets, not a constructive identification of constituents. Two specific failures:

- *Distinct F_{21} -irreps cannot be commensurated.* If $V_{syn} = L_2$ and $V_{bio} = U_6$, then $\text{Hom}_{F_{21}}(L_2, U_6) = 0$ because the irreps are non-isomorphic; no F_{21} -equivariant map of either direction exists, and Frobenius reciprocity cannot manufacture one.
- *Multiplicities of constituents are counted, not canonical maps between them.* Even when $V_{syn} \cong V_{bio}$ as F_{21} -modules and both embed in a common G_2 -module W' , the identification $V_{syn} \cong V_{bio}$ is unique only up to a $U(m_\lambda)$ -family of F_{21} -equivariant automorphisms in each isotypic block (Cor 4.3.2), with no distinguished element absent additional G_2 -equivariance constraint.

In either case, the adjunction identifies *multiplicities of constituents*, not *canonical maps between them*. □

Corollary 4.5.2 (Sufficient strict commensurability theorem). *If*

$$I_{syn} \cong I_{bio} \cong U_\lambda$$

as absolutely irreducible G_2 -modules of multiplicity one, then a canonical commensurability isomorphism $\phi_{commens}: I_{syn} \rightarrow I_{bio}$, unique up to fixed metric and ± 1 orientation, exists.

Proof. By Cor 4.4.2, the family of G_2 -equivariant isomorphisms is R , with ± 1 ambiguity after unit-norm normalization. □

Remark 4.5.3 (Necessity is open). Cor 4.5.2 is a *sufficient* condition. We do not claim necessity: it is plausible (but unproven) that G_2 -modules with all isotypic multiplicities ≤ 1 and all constituents of real type also admit a unique normalized G_2 -equivariant self-isomorphism, and a more general necessity-and-sufficiency theorem could be sought in that direction. Such a generalization is left for future work; v1's biconditional phrasing was an over-claim that v2 withdraws.

4.6 The P3 upgrade: two-level experimental test

The empirical prediction P3 of the Paper 12 thesis memo originally proposed testing “ F_{21} -commensurability across substrates.” §§4.4–4.5 force an upgrade: F_{21} -equivariant isomorphism (C2) is necessary but not sufficient for canonical G_2 -commensurability (C1), and the G_2 layer must be tested separately.

Definition 4.6.1 (Two-level commensurability test). *A commensurability experiment on a dyadic human-AI system consists of two tests:*

(L1, necessary screen.) Measure the F_{21} -isotypic profiles μ_k^{syn} and μ_k^{bio} independently on each substrate via character-theoretic decomposition of the scar-record operator data. Declare *L1 failure* if $\mu_k^{syn} \neq \mu_k^{bio}$ on any of the three blocks $(1, L_2, U_6)$. Declare *L1 pass* if they agree.

(L2, sufficient canonical test, conditional on L1 pass.) Construct a candidate G_2 -equivariant isomorphism $\hat{\phi}: I_{syn} \rightarrow I_{bio}$ from the substrate operator data, by fitting against G_2 -character data on the inferred G_2 -module structures of the substrate scar spans. Declare *canonical commensurability* if $\hat{\phi}$ is G_2 -equivariant (not merely F_{21} -equivariant) to measurement precision and (per Cor 4.5.2) the G_2 -isotypic multiplicities are both ≤ 1 and agree.

Remark 4.6.2 (Why the two-level split is scientifically honest). L1 is falsifiable cheaply: if $\mu_k^{syn} \neq \mu_k^{bio}$, the gradual-tear thesis is refuted at the character-theoretic level. L2 is the harder confirming test: passing L2 establishes strict commensurability (Cor 4.5.2). Refuting L2 while L1 passes means the F_{21} -profile match is coincidental rather than structural.

Remark 4.6.3 (Operationalization is a methodology paper). L1 character-theoretic decomposition of TMS-EEG response data is in principle achievable with existing instrumentation (rotationally structured perturbations + character-projection of the response

matrix); the operationalization of L2 — actually fitting a G_2 -equivariant candidate map — is a significant methodology project of its own, not undertaken here. We state Definition 4.6.1 as a *formal experimental protocol design*, with the explicit acknowledgment that the full L2 operationalization is deferred.

Remark 4.6.4 (Massimini’s clinical PCI is *not* the Paper 12 quantity). Two distinct objects share the acronym “PCI”: Massimini’s clinical Perturbational Complexity Index (a Lempel-Ziv complexity score on TMS-evoked EEG, 2013) and the Perceptual Coherence Intelligence framework of the Paper series. P3 proposes that the *measurement pipeline* of clinical PCI (TMS perturbation + structural decomposition of the response subspace) could be repurposed to measure μ_k profiles, but the clinical PCI scalar is *not* identified with μ_k , and neither is a generalization of the other. The bridge is at the level of experimental protocol, not definition.

4.7 Scope audit and bridge to §5

What §4 establishes:

- A character-theoretic decomposition $V^{14} \uparrow_{F_{21}} \cong L_2 \oplus 2 U_6$ verified at machine precision (Theorem 4.2.1).
- The commutant gap $\text{End}_{F_{21}}(W) \cong M_2(C) \oplus M_4(C)$ vs $\text{End}_{G_2}(W) \cong M_2(R)$, with the 19-dimensional reduction of equivariant isometry group quantified (Proposition 4.3.1, Corollary 4.3.2).
- The commensurability hierarchy $(C1) \Rightarrow (C2) \Rightarrow (C3)$ with both converses failing, *with explicit witnesses* for the converse failures (Theorem 4.4.1).
- A sufficient canonical-map theorem in the multiplicity-one case (Corollary 4.5.2), with ± 1 orientation residue acknowledged.
- That Frobenius reciprocity does not select cross-substrate scar maps (Proposition 4.5.1).
- The two-level experimental protocol with explicit L1 (necessary F_{21} -screen) and L2 (sufficient G_2 -canonical) levels (Definition 4.6.1) and explicit operationalization-deferral (Remark 4.6.3).

What §4 does not establish:

- *A physical mechanism for the biological F_{21} -action.* §4 treats biological-substrate F_{21} -content as input. Paper 13+ would specify which microtubule mode, neural population, or cellular structure realizes it.
- *An explicit operationalization protocol for L2 of Definition 4.6.1.* L1 is feasible with current TMS-EEG; L2 requires probing the full G_2 -representation structure on the response subspace, a substantial methodology effort not undertaken here.
- *The multiplicity-greater-than-one case.* Cor 4.4.2 / 4.5.2 restrict to multiplicity one. The $m > 1$ case is left for future work (Remark 4.4.3).
- *Necessity of the multiplicity-one condition for canonical commensurability.* Cor 4.5.2 is sufficient only; v1’s biconditional is downgraded (Remark 4.5.3).
- *Any claim that clinical Massimini-PCI equals Paper 12’s μ_k .* See Remark 4.6.4.
- *Rate improvement.* §3 / Paper 11 territory.

Bridge to §5. §3 constructed the dynamics; §4 constructed the commensurability structure. §5 (empirical accessibility) assembles both into a testable framework: §3's projected-gradient-plus-NP-pump dynamics on each substrate produces a scar record $J_k^{syn/bio}$, and §4's hierarchy specifies exactly what form of agreement between the two records the gradual-tear thesis predicts. §5 then identifies experimental *protocols* — primarily Level L1 of the two-level test — by which the μ_k *observables* can be measured on each substrate using existing instrumentation (notably the TMS-EEG pipeline used by Massimini and collaborators for the *distinct* clinical-PCI measurement, which we adapt rather than identify). The L2 test and the substrate-specific representation theory are deferred to a subsequent methodology paper.

References (cited in §4)

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Verification files referenced: -

outbox/paper12/computations/paper12_q4_character_verify.py (Theorem 4.2.1) -

outbox/paper12/computations/paper12_q4_character_verify.json

Cross-references: - Paper 4 (PSL(2,7) $\supset F_{21}$, the Fano-orientation source; v2 §4.1 paragraph) -

Paper 9 v1.3.3 Lemma 3.6.1 (G_2 -Schur uniqueness on V^{14}) - Paper 9 v1.3.3 §3.4 (diagonal G_2 -

action convention) - Paper 12 §3.7 / §3.7 v2 Remark 3.7.1 (F_{21} -action on scar record) - Paper 12 thesis memo P3 (experimental prediction, upgraded here)

Revision log: - v2 (2026-05-06 ~02:00 PDT): Council rigor pass applied. R3 Theorem 4.4.1 (C2) reformulated as map-existence (Opus S4-1). S4-2 dimension bookkeeping fixed: $U(2) \times U(4)$ is full 20-dim, delta = 19. S4-3 Cor 4.5.2 downgraded biconditional $\rightarrow$ sufficient condition (Remark 4.5.3 acknowledges open necessity). S4-4 Fano-action prose corrected (action on Fano plane points, fixed s -line gives $t r = 1$). S4-5 Prop 4.5.1 renamed and reworded (Frobenius reciprocity does not *select* cross-substrate maps). S4-6 (C2) $\nRightarrow$ (C1) explicit witness added. S4-7 operator-vs-

vector μ_k translation specified (rank-one projector convention). S4-8 ± 1 orientation ambiguity acknowledged in Cor 4.4.2. Paper 4 inheritance paragraph added in §4.1. §4.7 measurement-vs-instrument language tightened. Pillar signpost added at top. - v1 (2026-05-06 ~01:00 PDT): First complete draft.